

EDUSAMBAM ASSESSMENT SERIES

Practice Test: Right-Angled Trigonometry & Sine/Cosine Rule

PRINTABLE PRACTICE WORKSHEET

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Subject	Mathematics (Maths)	Topic	Practice Test: Right-Angled Trigonometry & Sine/Cosine Rule	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20

Total	45	—	100
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Section A — Multiple Choice

30 marks

1. The hypotenuse of a right-angled triangle is always:

Basic

- A. The longest side, opposite the right angle
 B. The shortest side
 C. Always horizontal
 D. Different depending on the angle used

Answer: ☐ A ☐ B ☐ C ☐ D

2. $\sin \theta$ is defined as:

Basic

- A. $\text{Adjacent} \div \text{hypotenuse}$
 B. $\text{Opposite} \div \text{adjacent}$
 C. $\text{Opposite} \div \text{hypotenuse}$
 D. $\text{Hypotenuse} \div \text{opposite}$

Answer: ☐ A ☐ B ☐ C ☐ D

3. $\tan \theta$ is defined as:

Basic

- A. $\text{Adjacent} \div \text{hypotenuse}$
 B. $\text{Opposite} \div \text{adjacent}$
 C. $\text{Opposite} \div \text{hypotenuse}$
 D. $\text{Hypotenuse} \div \text{adjacent}$

Answer: ☐ A ☐ B ☐ C ☐ D

4. A 5 m ladder makes a 60° angle with the ground. The height it reaches up a wall is closest to:

Intermediate

- A. 2.50 m
 B. 5.00 m
 C. 8.66 m
 D. 4.33 m

Answer: ☐ A ☐ B ☐ C ☐ D

5. A right triangle has opposite = 6 cm and adjacent = 8 cm. The angle θ is closest to:

Intermediate

- A. 53.1°
 B. 36.9°
 C. 45.0°
 D. 30.0°

Answer: ☐ A ☐ B ☐ C ☐ D

6. $\sin 30^\circ$ exactly equals:

Basic

- A. $\sqrt{3}/2$
 B. 1
 C. 0
 D. $1/2$

Answer: ☐ A ☐ B ☐ C ☐ D

7. The angle of elevation is measured:

Basic

- A. Downward from the horizontal
 B. From the vertical only
 C. Upward from the horizontal
 D. Always exactly 90°

Answer: ☐ A ☐ B ☐ C ☐ D

8. The sine rule states:

Basic

- A. $a/\sin A = b/\sin B = c/\sin C$
 B. $a^2 = b^2 + c^2 - 2bc \cos A$
 C. $\text{Area} = \frac{1}{2}ab \sin C$
 D. $a^2 + b^2 = c^2$

Answer: ☐ A ☐ B ☐ C ☐ D

9. The cosine rule should be used when you know:

Intermediate

- A. Two angles and a side only
 B. Only one side
 C. Nothing about the triangle
 D. Two sides and the included angle, or all three sides

Answer: ☐ A ☐ B ☐ C ☐ D

10. The formula for the area of a triangle using two sides and the included angle is:

Intermediate

- A. $\frac{1}{2} \text{ base} \times \text{height}$ only
 B. $\frac{1}{2}ab \sin C$
 C. abc
 D. $a^2 + b^2$

Answer: ☐ A ☐ B ☐ C ☐ D

11. Triangle ABC has $a = 7$ cm, $b = 9$ cm, and angle $B = 72^\circ$. Angle A is closest to:

Advanced

- A. 47.7° B. 72.0°
C. 90.0° D. 132.3°

Answer: ☐ A ☐ B ☐ C ☐ D

12. The "ambiguous case" in trigonometry can occur when using:

Advanced

- A. The cosine rule for a side B. The area formula
C. The sine rule to find an angle from SSA data D. Pythagoras' theorem

Answer: ☐ A ☐ B ☐ C ☐ D

13. A triangle has sides of 6 cm and 9 cm with an included angle of 50° . Its area is closest to:

Advanced

- A. 27.0 cm^2 B. 54.0 cm^2
C. 20.7 cm^2 D. 41.4 cm^2

Answer: ☐ A ☐ B ☐ C ☐ D

14. The cosine rule formula is:

Intermediate

- A. $a^2 = b^2 + c^2 - 2bc \cos A$ B. $a/\sin A = b/\sin B$
C. Area = $\frac{1}{2}ab \sin C$ D. $a^2 + b^2 = c^2$

Answer: ☐ A ☐ B ☐ C ☐ D

15. A triangle has sides $a = 5$, $b = 7$, $c = 10$. The angle C (opposite the longest side) will be:

Advanced

- A. Acute B. Exactly 90°
C. Impossible to determine D. Obtuse

Answer: ☐ A ☐ B ☐ C ☐ D

Section B — True / False

20 marks

16. The hypotenuse label never changes, regardless of which angle is being used.

Basic

☐ True ☐ False

17. $\cos \theta = \text{opposite} \div \text{hypotenuse}$.

Basic

☐ True ☐ False

18. To find a missing angle from two known sides, you should use an inverse trig function.

Intermediate

☐ True ☐ False

19. $\tan 45^\circ$ exactly equals 1.

Basic

☐ True ☐ False

20. The angle of elevation and the angle of depression between two observers looking directly at each other are always equal.

Intermediate

☐ True ☐ False

21. The sine rule and cosine rule are needed for triangles that do not necessarily have a right angle.

Basic

☐ True ☐ False

22. In standard triangle labelling, side a is adjacent to angle A.

Intermediate

☐ True ☐ False

23. The cosine rule can be used to find an angle when all three sides of a triangle are known.

Intermediate

☐ True ☐ False

24. A negative value inside $\cos^{-1}$ during the cosine rule always means a calculation error has occurred.

Advanced

☐ True ☐ False

25. In the $\frac{1}{2}ab \sin C$ formula, C must be the angle included between sides a and b.

Intermediate

☐ True ☐ False

Section C — Fill in the Blank

20 marks

26. The mnemonic for the three trigonometric ratios is SOH-CAH-_____.

Answer:

27. $\cos \theta = \text{adjacent} \div$ _____.

Answer:

28. To find a missing side using an angle and the hypotenuse, $\text{opposite} = \text{hypotenuse} \times$ _____.

Answer:

29. $\sin 90^\circ$ exactly equals _____.

Answer:

30. The angle measured downward from the horizontal to a point below the observer is called the angle of _____.

Answer:

31. In standard triangle labelling, side a is opposite angle _____.

Answer:

32. The area formula using two sides and the included angle is $\text{Area} = \frac{1}{2}ab \sin$ _____.

Answer:

33. The cosine rule formula is $a^2 = b^2 + c^2 - 2bc \cos$ _____.

Answer:

34. If a calculator gives an angle of 40° in an ambiguous sine rule case, the second possible value is _____ degrees.

Answer:

35. When two sides and the included angle are known (SAS), you should use the _____ rule.

Answer:

Section D — Column Matching

10 marks

36–40. Match each term to its correct definition.

Column A	Ans.	Column B
SOH	_____	A. $\tan \theta = \text{opposite} \div \text{adjacent}$
CAH	_____	B. $\sin \theta = \text{opposite} \div \text{hypotenuse}$
TOA	_____	C. $a^2 = b^2 + c^2 - 2bc \cos A$
Sine rule	_____	D. $\cos \theta = \text{adjacent} \div \text{hypotenuse}$
Cosine rule	_____	E. $a/\sin A = b/\sin B = c/\sin C$

Section E — Short Answer

20 marks

41. Explain how to decide which trigonometric ratio (sin, cos or tan) to use in a right-angled triangle problem.

42. Describe what an angle of elevation is.

43. Explain when you should use the cosine rule instead of the sine rule.

44. Describe what the "ambiguous case" of the sine rule means.

45. Explain how to find the area of a triangle when you know two sides and the angle between them.

Candidate Signature: _____

Invigilator Signature: _____

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Marks Obtained: _____ / 100 Checked By: _____

Result / Grade: _____ Date Checked: _____

Practice Test: Right-Angled Trigonometry & Sine/Cosine Rule — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. A | |
| 2. C | 9. D |
| 3. B | 10. B |
| 4. D | 11. A |
| 5. B | 12. C |
| 6. D | 13. C |
| 7. C | 14. A |
| 8. A | 15. D |

Section B — True / False

- | | |
|-----------|-----------|
| 16. True | 21. True |
| 17. False | 22. False |
| 18. True | 23. True |
| 19. True | 24. False |
| 20. True | 25. True |

Section C — Fill in the Blank

- | | |
|----------------|------------|
| 26. TOA | 31. A |
| 27. hypotenuse | 32. C |
| 28. sin | 33. A |
| 29. 1 | 34. 140 |
| 30. depression | 35. cosine |

Section D — Column Matching

- | | |
|-------------|---------------------|
| 36. SOH — B | |
| 37. CAH — D | 39. Sine rule — E |
| 38. TOA — A | 40. Cosine rule — C |

Section E — Short Answer (Model Answer Keywords)

41. opposite; adjacent; hypotenuse; known sides
42. upward; horizontal; above
43. two sides; included angle; three sides; SAS; SSS
44. two possible; angles; 180 minus
45. half; ab sin; included angle